ShiftSense

Continuous workforce control for SLA-driven operations — EXL x IIM Lucknow Live Action Project

Supratik Final solution · concept frozen · 27 September 2026

Predict demand. Plan capacity. Detect risk. Recover fast.

Core promise: keep the right skills in the right place at the right cost — even when reality changes.

Product definition: ShiftSense is a workforce operations decision layer that continuously compares demand with available skilled capacity, detects coverage risk, and recommends the least-disruptive feasible action to restore service within cost and policy constraints.

Positioning: ShiftSense is an **operational decision layer that sits on top of existing HRMS, attendance and contact-centre systems** — it integrates with them and does not replace them. **Fast, constraint-aware recovery is the differentiator; lower cost and lower deployment complexity are economic benefits.**

Structure of the approach

The document follows a product-management approach, from the market down to the build. Each section answers one question.

#	Section	Question it answers
0	Context	What did EXL ask for, and what do BPO, WFM and HCM mean?
—	Executive summary	What is ShiftSense, in one page?
1	Space	What market and operating environment are we in, and why now?
2	Competitors	Who solves this today, and where is the opening?
3	ICP and personas	Who exactly is the customer, and who uses the product?
4	Jobs to be done	What are those users trying to get done?
5	Old problems and current customer journey	How does it work today, and where does it break?
6	What changes	What does the new journey look like?
7	Proposed solution	What is the product, and how does it work?
8	Prioritization	What matters most, and what is in the MVP?
9	MVP	What is the smallest build that proves the idea, and how do we prove it?
10	Roadmap and guidelines	How does it go from PoC to product, and what rules guide the build?
11	Architecture	How is it built, and how does data flow?
12	Business impact and KPIs	How do we measure success and value?
13	Future potential, moat and risks	Why does it get stronger over time, and what could go wrong?
14	Template mapping and open questions	Where does each EXL template answer come from?

0. Context

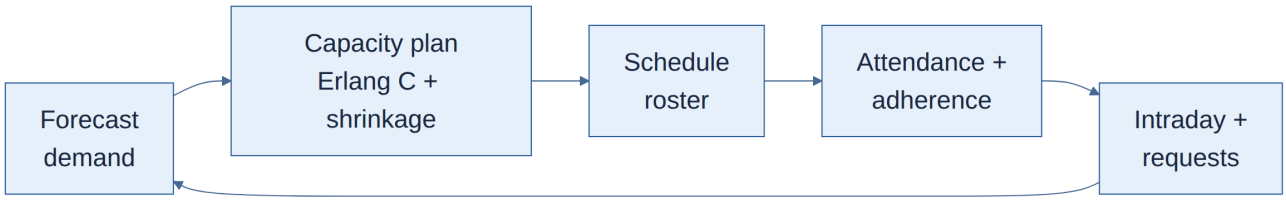
The brief

- **Programme:** EXL x IIM Lucknow Live Action Project (BizTech PM Club), problem statements shared on 31 Aug 2026.
- **Problem statement:** *"Managing shift scheduling, attendance, and workforce requests is manual, inefficient, and reliant on costly third-party tools. How might we optimize scheduling and unify core workforce management in one accessible platform?"*
- **Deliverables:** a functional prototype or PoC, and the completed Innovation & Solution Assessment template.

BPO, WFM and HCM in brief

A **BPO (Business Process Outsourcing)** firm such as EXL runs operations for clients from delivery centres staffed by large shift-based teams. It essentially sells **people's time, delivered on schedule**, against service levels (SLAs) such as *80% of calls answered within 20 seconds*. Too few people breaks the SLA; too many wastes cost.

WFM (Workforce Management) is the discipline of matching staff to demand interval by interval: forecast → capacity plan → schedule → attendance → intraday changes. **HCM / HRMS** is the HR system of record (employee master, payroll, leave). HCM answers *who works here*; WFM answers *who should work when*. ShiftSense sits on top of the HRMS, never in place of it.



The classic WFM cycle

Key terms

Term	Meaning
SLA / service level	The response target the client pays for, e.g. 80% of calls within 20 seconds
Interval	The planning slot, usually 15 or 30 minutes
AHT	Average handle time per contact; volume × AHT = workload
Erlang C	Queueing model that turns workload into agents needed to hit the SLA
Shrinkage	Paid time agents aren't available (breaks, leave, training, absence), typically 25–35%
Adherence	How closely an agent followed their schedule
Coverage	Available heads versus required heads, per queue and interval
Time-to-Cover	Minutes from detecting a staffing gap to restoring acceptable coverage
Open-shift offer vs marketplace	Offer: a TL offers an eligible, available shift to qualified agents (MVP). Marketplace: a self-service exchange where employees browse and claim shifts (future)
Freeze window	Period before a shift in which the optimizer's freedom to change it is limited

Term	Meaning
ICP / JTBD / MVP	Ideal customer profile / jobs to be done / minimum viable product
RICE / MoSCoW	Prioritization score (reach × impact × confidence ÷ effort) / Must–Should–Could–Won't

Executive summary

Workforce plans become obsolete as demand and workforce availability change. ShiftSense closes that gap by continuously comparing required skilled capacity with actual workforce availability. It has three connected capabilities:

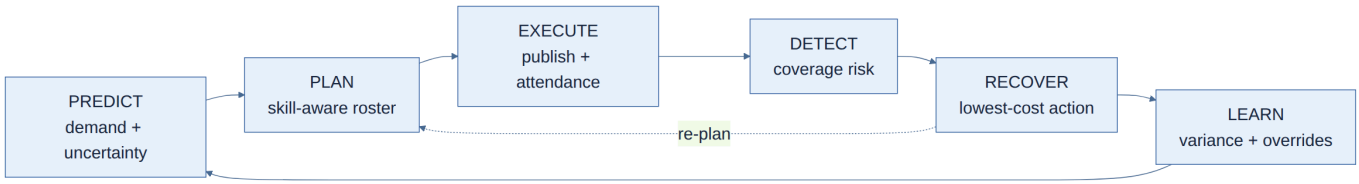
- **Plan:** forecast demand, calculate staffing requirements and generate a skill-aware, constraint-compliant and stable roster.
- **Detect:** monitor attendance, demand and adherence to find coverage gaps, and prioritize them by operational risk rather than headcount alone.
- **Recover:** generate feasible interventions (cross-skill redeployment, swaps, open-shift offers, overtime); simulate their coverage, cost and disruption; rank them; and let the team lead approve or modify. The affected slice of the schedule is then re-optimized locally, and the event, decision and outcome are recorded for audit and learning.

ShiftSense does not replace the HRMS, attendance system or contact-centre platform. It is the operational decision layer between workforce reality and workforce action. Fast, constraint-aware recovery is the differentiator; lower cost and lower deployment complexity are economic benefits.

The MVP proves one question: can ShiftSense create a better workforce plan, and restore coverage faster and at lower incremental cost than a credible manual process, without violating hard workforce constraints? All impact figures are hypotheses until tested.

The solution at a glance

Aspect	ShiftSense's answer
Intellectual structure	Demand → skilled capacity → plan → actual → risk → recovery → outcome
Product differentiation	Continuous, constraint-aware recovery
Core technical component	Skill-aware global optimization + local re-optimization
UX principle	Exception → recommendation → explanation → human decision
Headline business metrics	Time-to-Cover and recovery cost
Long-term moat	Organizational memory built from decisions and outcomes
Biggest validation question	Can the system consistently make better recovery decisions than an experienced person with the same information?



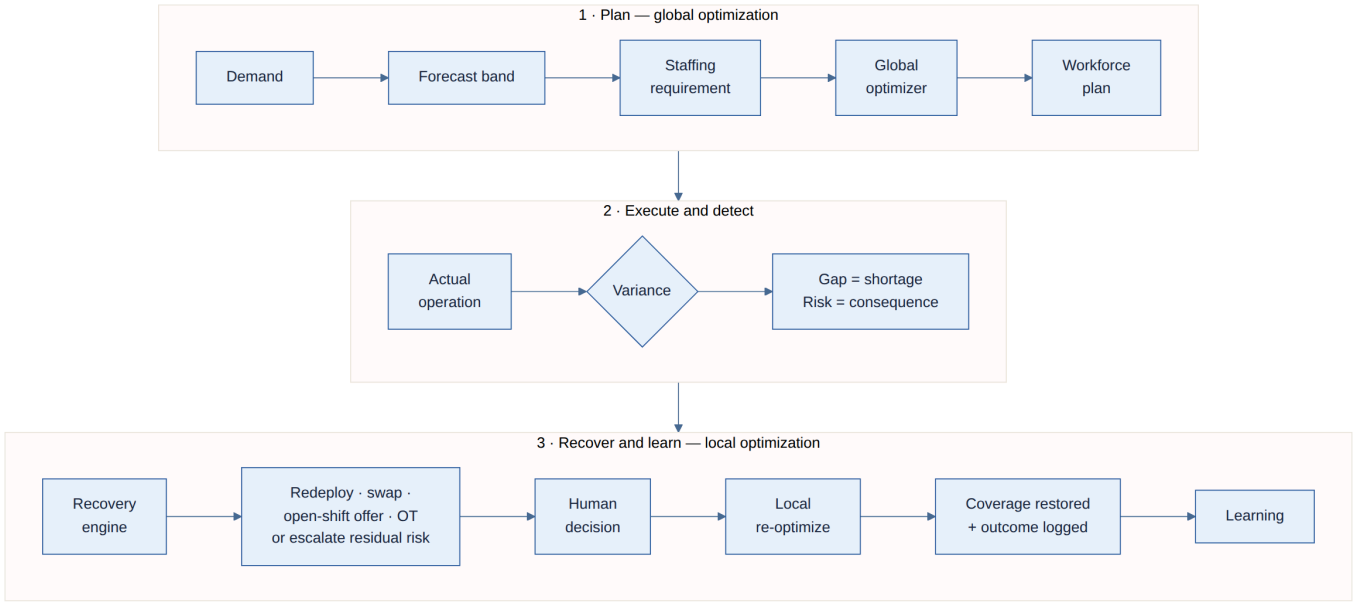
The ShiftSense control loop

What ShiftSense is — and is not

ShiftSense is NOT	ShiftSense IS
An HRMS or payroll system	The operational decision layer between workforce reality and workforce action
An attendance system or a replacement for the ACD	A consumer of their data that decides what to do with the workforce
An autonomous manager	A recommender; humans accept, modify or reject every consequential action
A legal compliance engine	An enforcer of organization-configured policies that flags potential violations
A generic chatbot	Deterministic engines with a natural-language interface on top
An absence-prediction product (yet)	A system that first proves what to do *after* someone is absent

The core decision flow

This flow is the core intellectual property of the proposal — more than any screen.



Core decision flow: global planning, detection, local recovery

1. Space

Workforce management is a roughly USD 10B software market growing 4–8% a year, and BPO delivery centres are its hardest operating environment. Vendors themselves name high implementation and licence cost as the main barrier to adoption.

Signal	Figure	Source
Global WFM software market, 2026	USD 9.76B, rising to USD 12.04B by 2031 (4.3% CAGR)	Mordor Intelligence
Asia-Pacific growth	6.17% CAGR, the fastest region	Mordor Intelligence
Incremental growth 2025–29	+USD 3.67B, 8.4% CAGR; “high implementation and maintenance cost” named as the key challenge	Technavio via PR Newswire

Where EXL sits: operations run 24x7 across many processes and clients, with interval-level SLAs, skill-based queues, high shrinkage and attrition, and state-specific labour rules (for example night-shift and weekly-off rules under Shops & Establishments Acts).

Why now:

- LLMs make natural-language requests and plain-English explanations practical.
- Hybrid and remote work make attendance harder to track and schedules more fluid.
- Margin pressure makes emergency overtime and SLA misses visible cost lines.

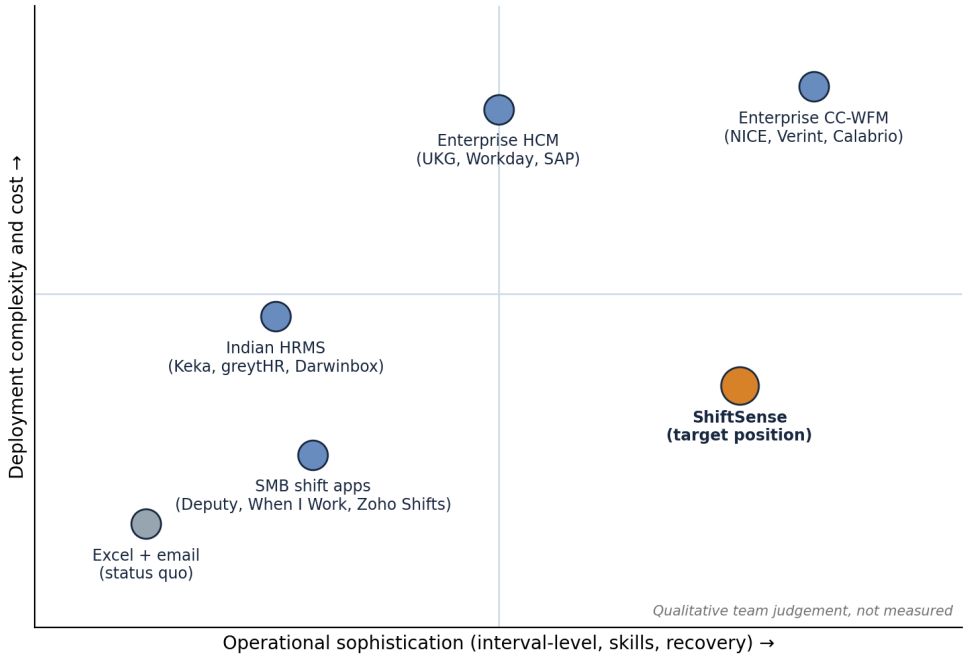
Market size supports the opportunity; it does not prove the product. The proof is in sections 9 and 12.

2. Competitors

ShiftSense is positioned as a decision layer on top of these systems, not as a replacement WFM or HCM platform.

The opportunity sits at the intersection of contact-centre-grade optimization, operational request and recovery workflows, and lower-complexity deployment. We do not claim competitors lack capabilities we have not verified.

Solution	Tier	Key features	Price (public)	Gaps for a BPO like EXL (as we understand them)	How ShiftSense differs
NICE (IEX WFM), Verint, Calabrio, Genesys WEM	Enterprise CC-WFM	Erlang forecasting, schedule optimization, adherence tied to the ACD	Quote-only, per seat, plus implementation	Quote-based enterprise pricing; higher apparent deployment complexity; analyst-centric UX	Supervisor-first recovery and requests; lighter deployment
UKG (Kronos), Workday, SAP, Oracle	Enterprise HCM	Time and attendance, payroll, HR	Quote-only	Scheduling is not their core; heavy implementation	Integrates with them; adds interval-level control
Deputy	SMB shift app	Auto-scheduling, timesheets, leave, swaps, forecasting	$5–$9 per user/month	Built for retail and hourly teams rather than interval SLAs and skill queues	SLA- and skill-aware coverage; recovery
When I Work	SMB shift app	Auto-scheduling, open shifts, swaps, messaging	$2.50–$8 per user/month	Same focus as Deputy	Optimization with stability, fairness and explanations
Zoho Shifts	SMB shift app	Scheduling, swaps, skills, time tracking	Free up to 25 users; from $1 per user/month	Manual rostering; no demand forecasting	Forecast-driven, skill-matched roster
Keka, greytHR, Darwinbox	Indian HRMS	Shift rosters, attendance, leave, payroll	Quote-based	Rosters built manually; no coverage view	Coverage-aware decisions on top of HRMS data
Excel + email + biometric export	Status quo	Flexible, free	Staff time	Manual, error-prone, no single view or audit trail	One operational view with a decision trail



Positioning map

Opening: high operational sophistication (interval-level, skills, recovery) at lower deployment complexity and cost — the lower-right of the map. *Enterprise vendors don't publish prices; we describe them as quote-based and do not assert their cost unless EXL shares actual figures.*

3. ICP and personas

ICP: a BPO, KPO or shared-services operation with 500–50,000 shift-based staff, bound by SLAs, using spreadsheets, an HRMS or a costly WFM suite. EXL's delivery centres are the first customer; mid-size Indian BPOs and GCCs come next.

Initial wedge (where value is strongest): a single BPO process with 24x7 operations, interval-level demand, multiple skills, meaningful shrinkage, high overtime, frequent intraday changes, an existing HRMS, and a manual or expensive WFM layer.

Persona	Role in the loop	Top pain today	ShiftSense proposition
WFM analyst	Plans	Days in Excel, rebuilds after every change	"Removes repetitive planning so you supervise the optimizer and run scenarios."
Team lead	Acts	Approves blind, fills gaps by phone	"See the consequence of every decision and the lowest-risk action; you stay in control."
Agent	Manages own work	Late rosters, lost requests, unexplained shifts	"Know your schedule. Request changes. Understand why decisions were made."
Ops / delivery head	Controls outcomes	No live view; overtime and penalties surprise them	"One operational view of where coverage will fail and what it costs to fix."
HR / compliance	Governs policy	Breaches found late, no audit trail	"Your policies applied in every plan and recovery, with a full decision record."

The **economic buyer** is the ops head (SLA and cost). The **champion** is the WFM analyst. **Daily users** are agents and team leads, so adoption depends on the mobile experience.

4. Jobs to be done

The core job: **“Keep the right number of people with the right skills in each interval, at acceptable cost, within policy — and keep it that way as things change.”**

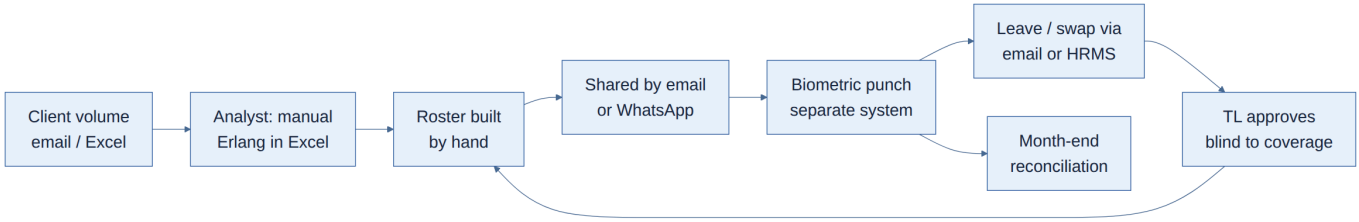
Persona	When…	I want to…	So I can…	Type
WFM analyst	next week’s forecast lands	generate a policy-compliant, skill-matched roster in minutes	stop rebuilding in Excel	Functional
WFM analyst	a client ramps or changes SLA	run what-if scenarios	advise ops before the change	Functional
Team lead	agents are absent at shift start	get ranked recovery options	restore coverage in minutes	Functional
Team lead	an agent asks for leave	see the coverage impact and alternatives	approve confidently	Functional
Agent	my plans change	request leave or a swap and see options	stay in control without chasing anyone	Functional + emotional
Agent	rosters come out	see nights and weekends shared fairly and explained	trust the system	Emotional / social
Ops head	the day is running	see only exceptions that need action	protect the SLA and the budget	Functional
HR	an audit comes	show every decision and the policy it met	avoid penalties and disputes	Functional

5. Old problems and current customer journey

Manual scheduling, fragmented tools and costly licences are symptoms. The root problem is that the plan becomes obsolete within hours, and recovery is slow, manual and invisible.

Illustrative week: on Monday the forecast needs 100 people and the roster assigns 102. By Tuesday three people are on leave, two are absent, volume runs 8% above forecast and one skill queue spikes. The original roster no longer answers the only question that matters: *what should operations do now?*

Where operational leakage occurs: forecast error + shrinkage + absence + skill mismatch + manual intervention + slow recovery = **SLA risk + overtime + idle capacity.**



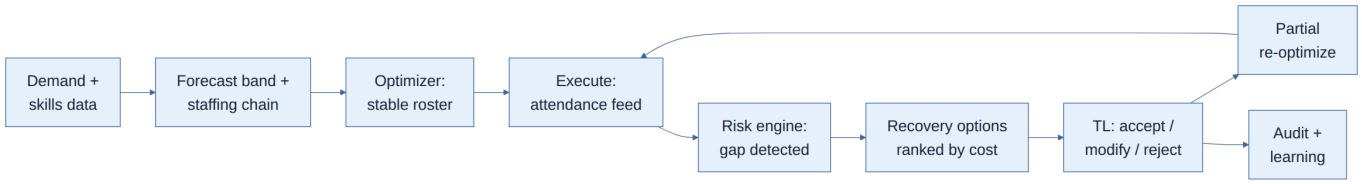
Current (as-is) journey

Stage	What happens today	Pain	Consequence
Forecast	Analyst copies volumes into Excel, applies Erlang manually	Slow, error-prone, person-dependent	Under- or overstaffing
Roster build	Hand-built grid, rules checked by eye	Hours to days; fairness disputes	Analyst time, attrition, policy breaches
Publish	Email or images	Late, conflicting versions	No-shows
Attendance	Biometric or VPN logs in a separate tool	Not linked to the plan	Gaps found after the fact
Requests	Leave in HRMS, swaps on chat	No coverage check	Understaffed intervals
Intraday change	TL phones around	Reactive, slow	Emergency overtime, SLA misses
Reporting	Month-end reconciliation	No live view	Management blind spots

These are typical BPO patterns, to be validated with the EXL expert; they are not EXL data.

6. What changes (to-be journey)

The weekly relay becomes a closed control loop: every change — a leave request, an absence, a volume spike — is detected, turned into ranked recovery options, decided by a human, and written back into the plan. The mindset shifts from *"build the roster once a week"* to *"keep coverage right continuously."*



To-be journey

Stage	Before	After
Forecast	Manual Excel Erlang	Forecast band per interval, converted through the staffing chain
Plan	Hand-built	Skill-aware, stable, explained roster in minutes
Execute	Separate attendance export	Planned vs actual linked live
Detect	Found when calls queue	Coverage risk flagged ahead of the interval
Recover	Phone calls, emergency OT	Ranked options by cost and disruption, one-tap approval
Requests	Blind approvals	Green / Amber / Red with alternatives
Learn	Month-end reconciliation	Variance and override reasons feed forecasts and policies

Language shift used throughout: one platform → **one operational view**; auto-roster → **demand-aware workforce planning**; leave approval → **coverage-aware workforce requests**; gap filler → **workforce recovery**; dashboard → **operations control tower**; LLM assistant → **natural-language workforce interface**.

7. Proposed solution

Thesis: ShiftSense continuously aligns workforce capacity with operational demand, balancing SLA, cost, policy and employee constraints as conditions change.

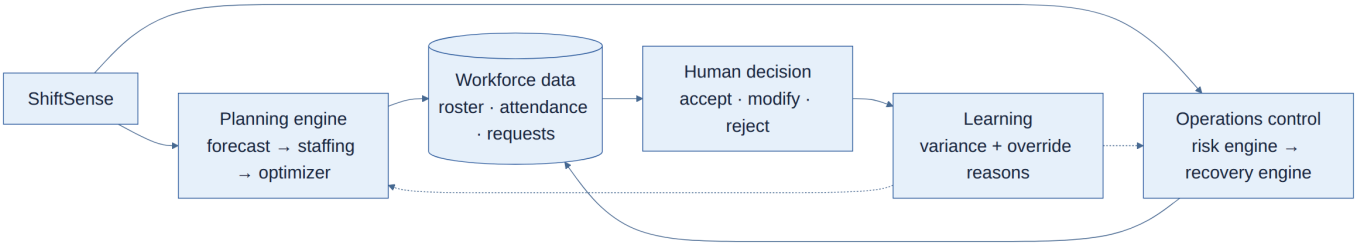
7.1 Three products in one loop

Product	Question it answers	Inputs	Output
A. Planning	What should next week's workforce plan be?	Forecast, skills, availability, policies, preferences	Optimized, stable roster
B. Monitoring	Is the operation on track right now?	Attendance, actual volume, adherence, staffing	Coverage risk by queue and interval
C. Recovery	We will miss coverage — what should we do?	Absence, spikes, leave, queue imbalance	Ranked feasible actions (redeploy, swap, open-shift offer, extension / OT) or residual-risk escalation

Existing WFM products address planning and monitoring; **ShiftSense focuses on the operational decision loop between planned and actual workforce capacity. Planning creates the baseline; recovery is the differentiating capability. Fast, constraint-aware recovery is the differentiator; lower cost and lower deployment complexity are economic benefits.** Two sequences are used deliberately:

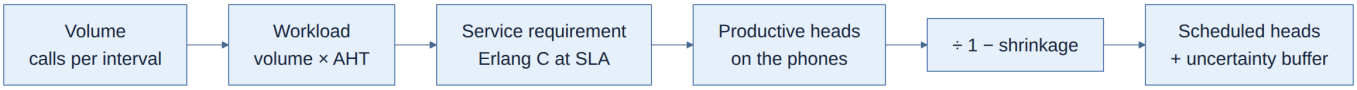
- **Product narrative:** Predict → Plan → Detect → Recover → Learn.
- **Decision hierarchy** (for every exception): Coverage → Risk → Recovery → Optimization → Execution.

	Planning optimizer	Recovery engine
Scope	**Global** — whole planning horizon	**Local** — the affected queue, day and people
Goal	Best feasible roster	Restore coverage with the **minimum necessary change** to the existing plan
Speed	Minutes, run before publishing	Seconds, run on each disruption
Freedom	Broad, outside freeze windows	Narrow, inside freeze windows



Product architecture

7.2 Demand → staffing chain



Staffing chain

Worked example: 130 calls in an interval → Erlang C at the target SLA → **18 = productive requirement** (agents on the phones) → 18 ÷ (1 − 0.30 shrinkage) ≈ **26 = scheduled requirement**.

Required heads are a **band, not a point** (e.g. 30–33). The buffer is **f(forecast uncertainty, SLA sensitivity, recovery flexibility)**: high uncertainty with few recovery options → larger buffer; high uncertainty with many cross-skilled replacements → smaller buffer. This links forecast, skills and recovery into one system and aims at minimum total cost at acceptable service, not maximum utilization. **MVP:** a transparent forecast-uncertainty band. **Future:** dynamically calibrated buffers learned from historical forecast error and operational outcomes. The model separates **forecast error** (demand was wrong) from **staffing error** (the plan or its execution was wrong). *Erlang C per queue is a standard approximation; multi-skill routing can be refined later with simulation. The PoC uses a transparent seasonal baseline — the story is decision quality, not model sophistication.*

7.3 Skill-based capacity

Staffing is not “30 employees”; it is **30 employees with the right capability mix**, from a skill matrix:

Agent	Claims · English	Claims · Hindi	Banking · English
A	1	1	0
B	1	0	1
C	0	1	1

Cross-skilled agents make **redeployment** possible: if Claims is short by 3 and Banking has 4 spare, two cross-skilled Banking agents can move before anyone is offered overtime.

7.4 The optimizer

Decision variable x(a,s,d) = 1 if agent *a* works shift *s* on day *d*. The objective:

$$\min \sum_{q,t}(c_u\,u_{q,t} + c_o\,o_{q,t}) + c_{ot}\sum_{a} OT_a + \lambda_f \,\text{Unfairness} + \lambda_p \,\text{PrefViolations} + \lambda_s \,\text{ScheduleChanges}$$

Planning objective

In plain English — minimize: understaffing risk + overstaffing cost + overtime + fairness violations + preference violations + unnecessary schedule changes. **Subject to:** skills, rest, leave, weekly hours, weekly offs, OT limits and organization policies.

- **Asymmetry:** c(u) > c(o). Understaffing risks SLA breach, backlog, burnout and OT escalation; overstaffing mainly costs idle labour.
- **Hard constraints:** one shift a day; minimum rest; maximum weekly hours; weekly offs; skill match; approved leave; organization-configured night-shift and OT limits.
- **Soft constraints:** fairness weighted by preferences and history of night and weekend burden; agent preferences; team continuity.
- **Stability:** a penalty on changes to already-published schedules prevents marginal cost improvements from causing disproportionate schedule disruption. *Principle: stability has economic and employee value; optimization should not trade it away for marginal gains.*

Freeze window (time before shift)	Optimizer freedom
More than 7 days	Full re-optimization
3–7 days	Limited changes, high stability penalty
Under 24 hours	Only operationally necessary changes
Current shift	Recommendation and redeployment mode only

7.5 Coverage risk engine

Coverage is a state; risk is the consequence. A coverage gap tells us what is missing; risk tells us whether the missing capacity actually requires intervention, and how urgently. Two equal gaps can carry very different risk.

Coverage state + forecast uncertainty + SLA sensitivity + skill scarcity + time to interval + recovery availability + intervention cost → operational risk → priority.

Risk combines the gap with forecast uncertainty, SLA sensitivity, queue volume, skill scarcity, time until the interval, available replacements and intervention cost (illustrative):

Level	Queue and interval	Required / available	Why	Recommended
Critical	Claims, 10:30–11:30	42 / 38 (gap −4)	High SLA sensitivity, few qualified replacements, 20 min to interval	Redeploy 2 cross-skilled; offer 2-hour OT to 2
At risk	Banking, 14:00–14:30	30 / 29 (gap −1)	4 qualified swap candidates	Offer a swap
Healthy	Others	≥ required	—	No action

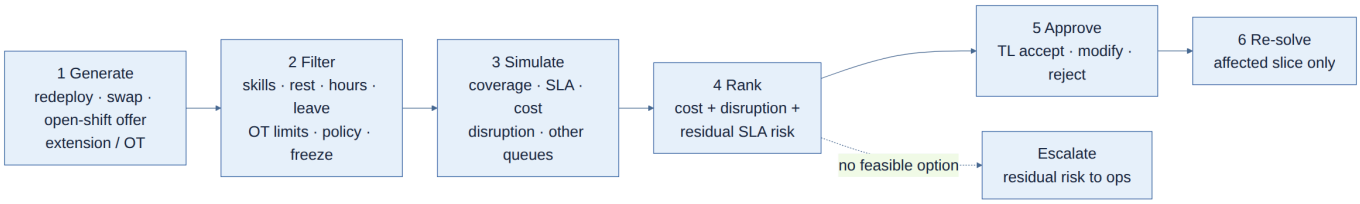
7.6 Recovery engine

Recovery selects the lowest-cost feasible intervention that restores required coverage while respecting SLA, skill, policy, stability and disruption constraints. It makes the **minimum number of schedule changes** needed:

$$\min \sum_k \text{cost}_k\, y_k \;+\; \lambda_b\, \text{BlastRadius}(y) \;+\; \lambda_r \text{ResidualGap}(y) \quad \text{s.t. skills, policies, freeze windows}$$

Recovery objective

The engine works in six steps — **Generate → Filter → Simulate → Rank → Approve → Re-solve**:



Recovery pipeline

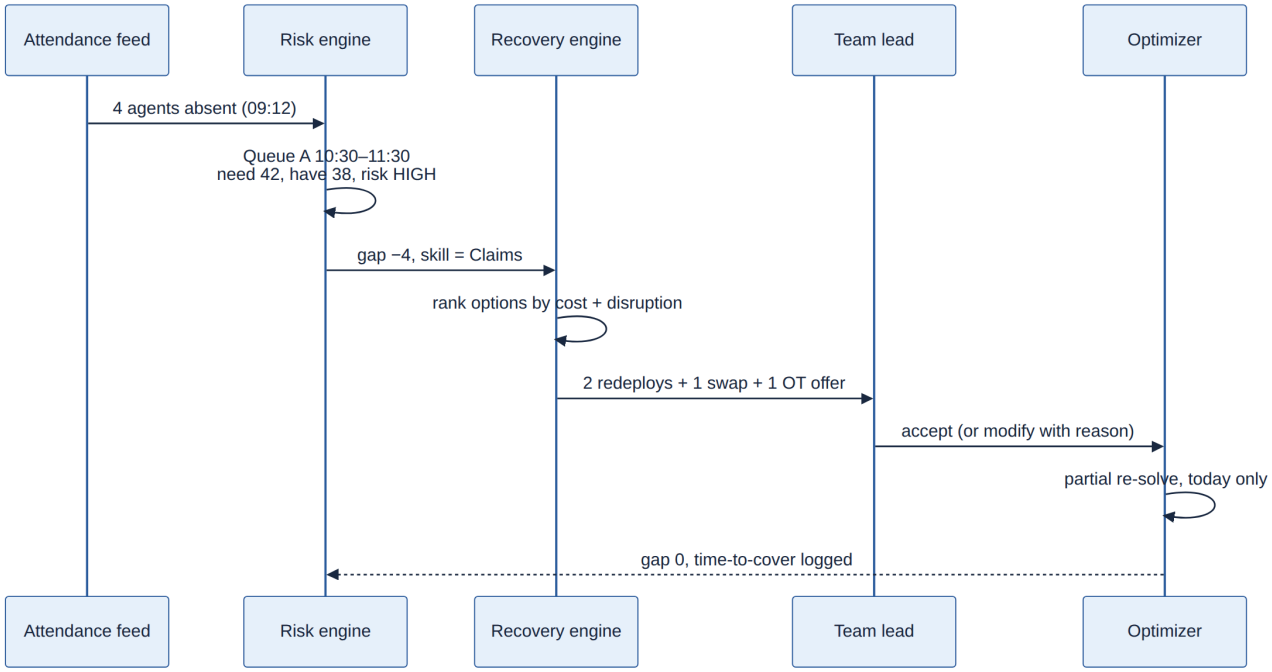
Candidate recovery actions: cross-skill redeployment → shift swap → open-shift offer → shift extension / OT → **escalate residual risk**. The first four are *recovery actions* — changes to workforce allocation the system can recommend. Escalation is not an action: it is what happens to the *residual risk* that remains when no feasible action fully closes the gap. The list is not a fixed cost order: a swap can be more disruptive than OT, and a redeployment can weaken the queue it comes from. Ranking uses **minimum feasible cost + disruption + residual SLA risk**.

Blast radius = the employees, shifts and queues a recovery touches. Option A (OT, ₹5,000, 2 agents affected) may beat option B (redeployment, ₹3,500, 11 agents affected, another queue weakened).

Residual-risk escalation: when no feasible action closes the gap, the system does not "accept risk" on its own; the human owns the trade-off. It escalates with the residual gap, affected interval, expected SLA exposure, remaining alternatives and their incremental cost: *"No feasible recovery. Residual exposure: −2 heads for 30 minutes, high SLA sensitivity. Ops decision required."*

Recommendation card (what the TL sees, illustrative):

Field	Content
Problem	Claims 10:30–11:30 · required 42, available 38, gap −4
Risk	Critical — high SLA sensitivity, 2 qualified replacements
Step 1	Redeploy agents A and B from Banking → gap −2
Step 2	2-hour OT for agents C and D → gap 0
Expected result	Coverage 38 → 42
Cost / disruption	₹X · blast radius 4 agents, 2 queues · medium
Confidence	High (5 candidates, forecast confidence high, policies complete)
Data as of	10:08
Why not the alternative?	OT-only was not recommended: two qualified cross-skilled agents were available at lower expected cost
Actions	Accept · Modify · Reject (reason required)

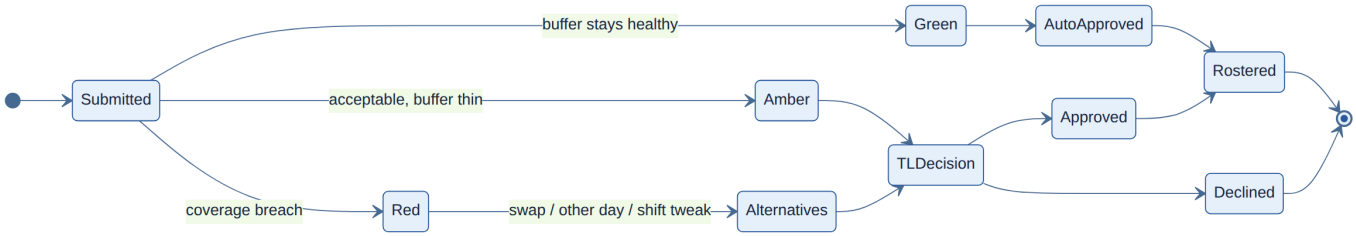


Disruption and recovery sequence

7.7 Coverage-aware requests

State	Condition	Action
Green	Coverage stays comfortably above requirement	Auto-approve under policy
Amber	Acceptable, but the buffer becomes thin	Team lead approves
Red	Approval would breach coverage	Offer alternatives, then team lead decides

Instead of *"Rejected — insufficient coverage"*, the agent sees alternatives: **swap with Rahul**, **take Thursday instead**, or **take Friday with a 2-hour shift adjustment** — each keeping coverage intact.



Request lifecycle

7.8 Operations control tower

The ops head's first question is *"what requires my decision?"*, so the home screen leads with actions, then aggregates (illustrative):

7 exceptions require action

- **Critical:** Claims −3 at 10:30 · Banking skill shortage at 11:00 · night-rule conflict for 2 agents
- **At risk:** absence cluster of 14 · OT threshold exceeded · 2 lower-priority gaps

Coverage 97.2% · OT 31 h · 18 absent · 4 unfilled shifts · data refreshed 2 min ago

7.9 Trust: explainability, override, audit and the NL interface

- **Explanations at three levels.** Agent: *"You got this shift because you're certified for Process A, demand is high, you prefer nights and your night burden is below team average."* TL: *"Leave declined: 2 qualified agents already absent, coverage would fall below threshold, no swap found."* Ops head: *"OT rose because of +8% forecast error, +4% absence and a Queue B skill shortage."*
- **Human in the loop, exactly as designed:** system detects → recommends → explains → TL or analyst decides → system records the reason → executes and re-solves → outcome recorded. No final workforce decision is automated in the MVP.
- **"Why not?" alongside "Why?"** Every recommendation states why the main alternative was not chosen, e.g. *"OT was not recommended because two cross-skilled agents were available and redeployment had lower expected cost and disruption."*
- **Override as a feature.** Every recommendation offers **Accept / Modify / Reject**, with a reason on modify or reject. Override reasons become product intelligence.
- **Recommendation confidence** = data freshness + forecast confidence + candidate availability + **policy certainty** (are the relevant rules configured and complete?).
- **Audit record** per decision: event → system state → recommendation → decision → outcome, with timestamps.
- **Natural-language interface.** The LLM is not the innovation; the pipeline is: **text → structured request → policy check → coverage simulation → optimization → human decision.** The LLM only parses and explains.

8. Prioritization

Features are grouped by their role in the product thesis, with a separate column for the much smaller MVP scope. Version 1's RICE scores were preliminary team estimates, not customer data, so they no longer decide strategy.

Tier	Feature	Product priority	In MVP?
Core	Demand forecast (baseline) + staffing chain	Must	Yes

Tier	Feature	Product priority	In MVP?
Core	Skill-aware optimizer with hard-constraint engine	Must	Yes
Core	Coverage monitoring + coverage-risk detection	Must	Yes
Core	Recovery recommendations + partial re-optimization	Must	Yes
Core	Leave / swap workflow (Green / Amber / Red)	Must	Yes
Core	Human accept / modify / reject + audit trail	Must	Yes
Core	Exception-first control tower (exceptions → risk → action → decision)	Must	Yes (lightweight)
Core	Transparent forecast-uncertainty band (staffing buffer)	Must	Yes (rule-based)
Enabler	Schedule stability + freeze windows	Should	Yes (penalty only)
Enabler	Cross-skill redeployment	Should	Yes
Enabler	Fairness optimization	Should	Yes (simple)
Enabler	Explanations incl. “why not”	Should	Yes (TL level)
Enabler	What-if scenarios (demand, absence, SLA change)	Should	Light version
Enabler	Open-shift marketplace (self-service; employees browse and claim shifts)	Should	No — the MVP has TL-issued open-shift offers only
Enabler	Acceptance and override analytics	Should	No
Future	Natural-language interface	Could	Demo only
Future	Dynamic / learned staffing buffers (from historical forecast error and operational outcomes)	Could	No
Future	ML demand forecasting, absence prediction	Could	No
Later	Native app, payroll, deep integrations, external SaaS, gig pool	Later	No

Ordering rules: table stakes (schedule, attendance, requests) must work; core differentiation (optimizer, coverage simulation, recovery) gets the most build time; absence prediction comes only after recovery is proven — first show what happens *after* someone is absent, then predict the absence.

9. MVP (minimum viable product)

The MVP question: can ShiftSense create a better workforce plan, and restore coverage faster and at lower incremental cost than a credible manual process, without violating hard workforce constraints? Nothing else

needs to be proven in the PoC.

MVP loop

1. **Input:** 50 agents with skill matrix, availability, preferences and contract rules; demand by 30-minute interval; SLA; AHT.
2. Forecast demand (transparent baseline, with a band).
3. Compute required heads through the staffing chain.
4. Generate the optimized, skill-matched roster.
5. Show coverage, cost, fairness, stability and violations.
6. **Inject a disruption:** 4 agents absent.
7. Detect coverage risk by queue and interval.
8. Recommend ranked recovery actions (redeploy, swap, open-shift offer, extension / OT), or escalate residual risk.
9. TL accepts, modifies or rejects with a reason.
10. Partially re-optimize; show coverage restored, Time-to-Cover and recovery cost.

A leave request with Green / Amber / Red handling and alternatives is included, because “workforce requests” is explicitly in EXL’s brief.

Out of MVP: real integrations, payroll, SSO, native mobile, multi-site, ML forecasting, absence prediction, policy-builder UI (policies are config files in the PoC).

A credible baseline

A random or greedy roster is not a credible “manual” baseline. The baseline is a **rule-based heuristic that mimics an analyst**, with its rules published in the demo appendix so no one can say it was made deliberately weak:

1. Satisfy hard constraints (rest, weekly hours, weekly offs, leave).
2. Fill required heads per interval, highest-demand intervals first.
3. Distribute skills so each queue meets its minimum.
4. Rotate nights and weekends in a simple round-robin for fairness.
5. Handle exceptions one at a time in the order they arrive.
6. Recover sequentially: first eligible agent on the same queue, then OT.

Ablation comparison (to be filled with PoC results)

Approach	Coverage	Understaffed intervals	OT hours	Fairness	Schedule changes	Build time	Time-to-Cover
Manual heuristic	—	—	—	—	—	—	—
Optimizer only	—	—	—	—	—	—	—
Optimizer + fairness + stability	—	—	—	—	—	—	—
ShiftSense full loop (with recovery)	—	—	—	—	—	—	—

What each layer should prove: optimizer → better coverage; + fairness → better distribution of nights and weekends; + stability → fewer schedule changes; + recovery → lower Time-to-Cover and recovery cost.

Hypotheses and gates

Hypothesis	Measurement (same synthetic dataset)	Gate to proceed
H1 Planning efficiency: roster preparation effort falls ≥ 70%	Build time, heuristic vs ShiftSense	≥ 70% without worse coverage or fairness
H2 Service: under-covered workload falls ≥ 20%	Under-covered workload ÷ total workload (normalized by demand)	≥ 20% at equal or lower paid hours
H3 Recovery speed: median simulated Time-to-Cover under 10 minutes	Injected disruption, timed end to end	Median under 10 minutes in simulation
H4 Recovery economics: equal coverage restored at lower incremental cost than OT-first recovery	Cost of chosen actions vs baseline	Lower cost at equal restored coverage
H5 Operational safety: zero hard-constraint violations	Automated check on every plan	Zero violations
H6 Adoption (validated in the pilot, not the PoC): TLs accept or lightly modify a high share of recommendations	Acceptance rate, modification rate, rejection reasons from real TLs	Majority accepted or lightly modified; rejection reasons show no systematic flaw

In the PoC, Time-to-Cover measures system decision latency plus a simulated approval; real human latency is measured only in the pilot.

Demo script (about 5 minutes)

Time	Scene	What the audience sees
0:00	**Plan**	Demand → staffing chain → skill-aware optimized roster, compared with the manual heuristic
0:45	**Break the plan**	“Four agents become unavailable.”
1:15	**Detect**	Claims: 42 required, 38 available, gap −4, **Critical**
1:45	**Diagnose**	High SLA sensitivity + limited qualified replacements + 20 min to interval
2:15	**Recover**	Redeploy, swap and OT options, each with cost, coverage and blast radius
3:00	**Decide**	TL accepts one action and modifies another with a reason
3:30	**Re-solve**	Gap −4 → 0; Time-to-Cover, recovery cost, schedule changes, SLA status
4:15	**Request**	An agent’s leave request goes Red; alternatives offered (other day, swap, shift adjustment)
4:45	**Close**	*“ShiftSense doesn’t just build the workforce plan. It keeps the operation covered when reality changes.”*

Recovery options shown at 2:15 (illustrative):

Option	Cost	Blast radius	Gap after
Redeploy 2 cross-skilled agents	₹X	2 agents, 2 queues	−2

Option	Cost	Blast radius	Gap after
+ Swap 1	₹X	2 agents	−1
+ OT 1	₹X	1 agent	0

10. Roadmap and guidelines

The roadmap is organized around validated value, not feature accumulation: each phase must prove one thing before the next begins.

Phase	Proves	Scope	Gate
0	Planning quality	PoC on synthetic data; plan + disrupt + recover	Beats the manual heuristic on H1–H5
1	Adoption	One process, one site, one team; CSV imports	Acceptance rate high enough to trust; sustained TL usage
2	Recovery	Live attendance feed; intraday recovery	Time-to-Cover and recovery cost improve on real disruptions
3	Economics	ROI calculator on real OT, pay and SLA data	Net benefit positive vs platform cost
4	Scale	Multi-site, multi-client, API integrations, Policy Builder	Consistent results across processes
5	Intelligence	Learned staffing buffers, absence risk, proactive recommendations	Measurable gain over rule-based recovery

Pilot design: a baseline period on the current process, then a ShiftSense period, on one process, one site and one team. Where possible, add a **control group** (a similar process that keeps the existing method) so improvements aren't just easier demand. Compare coverage, OT, roster time, TL interventions, adherence, schedule changes, SLA outcomes and request turnaround, **normalized by workload** — e.g. OT hours per 1,000 workload hours, under-covered workload ÷ total workload, TL interventions per 1,000 transactions, recovery cost per unit of recovered workload.

Build guidelines

- **Integrate, don't replace:** the HRMS stays the system of record for employees and payroll.
- **Humans decide:** the optimizer proposes; TLs and analysts accept, modify or reject, and every decision is explained and logged.
- **Rules as configuration, not code,** so each site or process has its own policy pack, validated by HR and legal.
- **Stability over marginal gains:** respect freeze windows; never churn published schedules for small improvements.
- **Mobile-first for agents and TLs** (responsive web / PWA, no native app in early phases); analyst tools stay desktop.
- **Augment first:** analysts become supervisors of the optimizer, not its victims.

Submission guidelines for the team

- Lead with the disruption-and-recovery demo; evaluators should see the loop working in the first minutes.
- **Scope is frozen.** No absence prediction, ML forecasting, chatbot features, native app, payroll, deep integrations, gig marketplace, complex policy builder or multi-site optimization until the PoC proves the loop.

- Show numbers versus the manual heuristic on the same data, not claims.
- Keep scope tight: scheduling, attendance, requests, recovery. Payroll and hiring are out.
- Label every figure as sourced, simulated or hypothesis.

11. Architecture

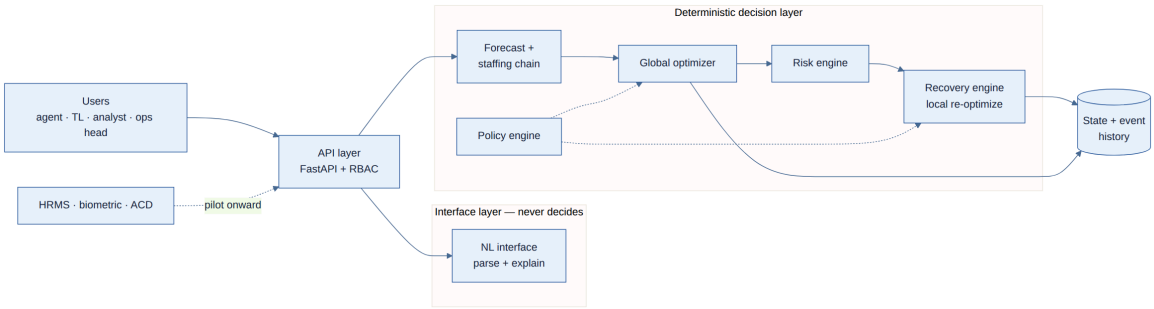
ShiftSense is a decision layer on top of the HRMS, attendance and contact-centre systems. A web front end, a Python API and a deterministic decision layer (forecast, global optimizer, risk, recovery, policy) over one store that keeps current state and event history. A separate interface layer (the LLM) only parses and explains — AI does not make the operational decision.

Principles and system-of-record boundaries

HRMS says *who the employee is*; attendance says *what happened*; ACD says *what the demand was*; ShiftSense decides *what to do with the workforce*. ShiftSense enforces organization-configured policies and flags potential violations; it does not guarantee legal compliance.

Data	System of record
Employee master, payroll	HRMS
Attendance punches	Biometric / attendance system
Contact volume	ACD / contact-centre platform
Forecast, staffing requirement	ShiftSense
Roster, request workflow	ShiftSense
Risk, recovery and optimization decisions	ShiftSense
Decision audit trail	ShiftSense

Components



Solution architecture

Component	Responsibility	MVP tech	Scale-up tech
Front end	Role-based views; exception-first control tower	Streamlit or React	React / Next.js, PWA
API layer	Auth, RBAC, endpoints	FastAPI	Containers + API gateway
Forecast + staffing chain	Forecast band → required heads per queue	pandas + baseline + Erlang C	Prophet / LightGBM; simulation for multi-skill
Global optimizer	Weekly plan across the horizon, with stability	OR-Tools CP-SAT	Async jobs; decomposition by team or day
Risk engine	Coverage risk by queue and interval	Python rules + thresholds	Uncertainty-aware scoring

Component	Responsibility	MVP tech	Scale-up tech
Recovery engine	Local re-optimization: feasible actions ranked by cost, SLA impact, blast radius	CP-SAT on the affected slice	Learned costs from outcomes
Policy engine	Rules as configuration	YAML / JSON	Policy Builder UI per site
NL interface	Text → structured request; explanations	LLM API with JSON schema	Guardrails + logging
Store	Current state and event history	SQLite	PostgreSQL + event stream

Data flow

1. **Data in:** interval volumes and agent data (skills, contracts, leave) — CSV in the PoC; HRMS, ACD and biometric feeds later.
2. **Forecast → how many:** volume × AHT → Erlang C → productive heads → shrinkage → scheduled band. Example: 130 calls → 18 on the phones → 18 ÷ 0.7 ≈ 26 rostered.
3. **Optimizer → who, when:** reads requirements, agents and policies; writes a stable roster.
4. **Execution → what happened:** attendance is stored against the roster; adherence and live coverage are computed.
5. **Detect → recover:** an event (absence, leave request, spike) triggers the risk engine, then ranked recovery options, then a TL decision, then a partial re-solve.
6. **Learn and audit:** every event, recommendation, decision and outcome is logged; variance is attributed to forecast error, absence, adherence, skill mismatch or scheduling.

Target design is event-driven: AgentAbsent, LeaveRequested, SwapRequested, DemandSpike, SLAAtRisk, ShiftStarted, ShiftEnded → risk engine → optimizer / recovery → recommendation → human decision → outcome. The store records *when* things happened and *what the system knew then*.

12. Business impact and KPIs

North Star — Productive Coverage Rate: the percentage of demand intervals in which required skill coverage is met within the configured cost envelope.

Area	KPIs
Service	SLA attainment; under-covered workload; overstaffed workload; coverage buffer
Recovery	See the recovery KPI table below
Workforce	Adherence; **schedule stability**; fairness; preference satisfaction; absenteeism
Cost	Overtime hours; excess staffing; cost per productive hour; licence cost
Productivity	Roster build time; request turnaround; TL intervention time; analyst hours saved
Product quality	Forecast error; optimizer runtime; **acceptance and override rate (with reasons)**; **recommendation success rate**; **data freshness**

Recovery KPIs — a complete measurement system:

Recovery KPI	Answers
Time-to-Cover	How fast is coverage restored?
Decision latency	How fast does the system recommend (vs human approval delay)?
Recovery success rate	How often does the recommended action actually restore coverage?
Recovery cost	How expensive was the recovery?
Blast radius	How many employees, shifts and queues did it disturb?
Unresolved critical gaps	What remained and was escalated?

New metric definitions: *Time-to-Cover* — detection → coverage restored (e.g. manual 10:15 → 11:00, 45 min; ShiftSense 10:15 → 10:20, 5 min). *Decision latency* — detection → recommendation, separating system speed from approval delay. *Recovery cost* — cost of actions used to close a gap. *Recommendation success rate* — share of actions that actually restored coverage. *Data freshness* — age of the data behind each recommendation.

Economic value

The customer buys better operational decisions, not a cheaper WFM tool. The value chain: **better forecast → better staffing → less excess capacity → earlier risk detection → faster recovery → less OT and SLA leakage → less manual work → economic value.** The levers are ordered by operational importance; software savings come last, which also keeps the differentiation from becoming a price competition.

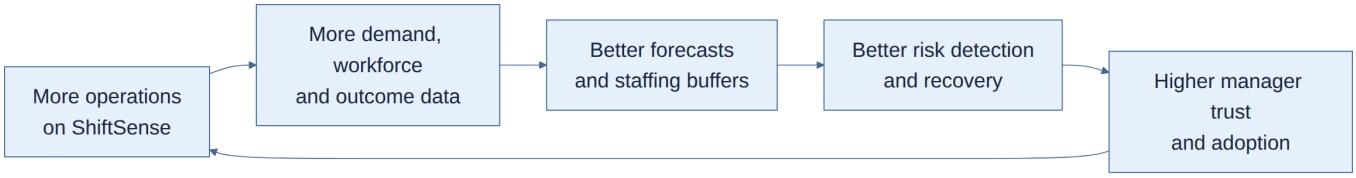
Lever	What changes	Measured by
1. SLA protection	Less under-covered workload	SLA attainment
2. Overtime reduction	Feasible alternatives considered before OT	OT hours and premium
3. Workforce utilization	Capacity matched to demand with a sized buffer	Cost per productive hour
4. Analyst and TL productivity	Less manual rostering and gap-filling	Analyst hours saved; TL intervention time
5. Schedule stability	Fewer unnecessary changes for employees	Changes per agent per week
6. Software savings	Less dependence on per-seat tools	Licence spend avoided

Annual operational benefit − platform cost = net economic value, computed in the pilot with EXL's actual seat counts, pay rates and OT data. Licence reference point: a mid-tier SMB tool at $6.50 per user a month (Deputy Core) would cost about $780,000 a year for 10,000 agents (illustrative). We describe *potential* SLA-penalty avoidance but make no rupee claim without EXL data.

13. Future potential, moat and risks

Moat: organizational memory

The solver is not the moat. The moat is accumulated **workforce data, operational policies, decision history, override feedback, customer-specific constraints, recovery intelligence and — most valuable — outcomes**: *"We recommended X; the manager chose Y; Y restored coverage in 6 minutes and cost ₹2,000 less."* That record is a true decision-learning loop. Over time ShiftSense can say: *"This process sees a 15% spike every Monday 10–12; the current roster has only a 3% buffer."*



Data flywheel

Scalability and commercial potential

- **Users and industries:** EXL processes first; then mid-size BPOs, GCCs and other SLA-driven shift operations (healthcare staffing, logistics, retail support).
- **Scale path:** decomposition by team or day keeps optimizer runtimes bounded; policy packs per site; multi-tenant SaaS only after economics are proven (phase 3).
- **Commercialization:** internal cost saving for EXL first; later an offering to EXL clients and other BPOs.

Risks, ranked

Rank	Risk	Mitigation
1	Data quality (wrong skills or availability invalidate plans)	Import checks; freshness shown on every recommendation; skill matrix owned by TLs
2	Manager trust	Explanations; accept / modify / reject; override analytics; augment-first rollout
3	Schedule instability	Stability penalty and freeze windows
4	Integration complexity	CSV first; APIs only after value is proven
5	Rule complexity	Policies as configuration; HR/legal validation; violations flagged
6	Forecast error	Forecast bands and buffers; separate forecast error from staffing error
7	Recommendation correctness (an action that looks right but does not restore coverage)	Every recommendation shows expected coverage impact, cost, assumptions and affected employees before approval; recommendation success rate tracked
8	Optimizer scalability	CP-SAT time limits; decomposition; partial re-solve
—	LLM misreads a request	Structured output, confirmation step, form fallback

14. Template mapping and open questions

EXL template section	Take from here
1. Problem statement & target users	Sections 0, 3, 4 and 5
2. Differentiation & innovation	Executive summary and section 7
3. Competitive landscape	Section 2
4. Solution workflow + diagram	Sections 6 and 7.6 (loop, to-be journey, recovery sequence)
5. Architecture + diagram	Section 11
6. Future potential & scalability	Sections 10, 12 and 13

Open questions for the EXL expert

- Which WFM tool and HRMS does EXL use today, and at what cost per seat?
- Typical interval volumes, AHT, SLA targets and shrinkage for one sample process?
- How are intraday gaps handled today, and how long does covering one usually take?
- Which policies (night shift, weekly off, rest hours, OT limits) matter most?
- Deadline, evaluation criteria and demo format?

Sources

- Mordor Intelligence — Workforce Management Software Market
- Technavio via PR Newswire — WFM market to grow by USD 3.67B (2025–2029)
- Deputy pricing · When I Work pricing · Zoho Shifts pricing · Keka pricing

Pricing pages accessed September 2026. Figures marked “illustrative” are not EXL data.